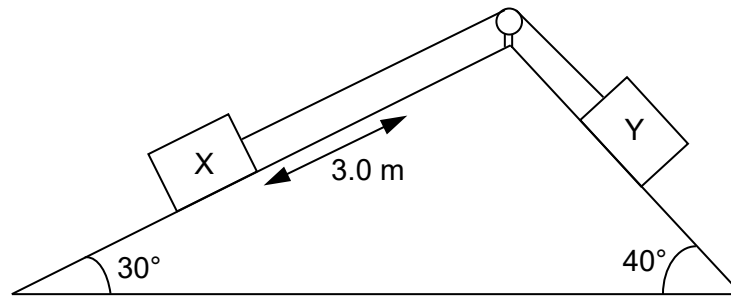


- 9 Two boxes X and Y of masses 2.0 kg and 3.0 kg respectively are connected by a light inextensible rope passing over a smooth pulley. X and Y are on smooth inclined planes.

X starts from rest and moves up the plane inclined at 30° to the horizontal.



What will be the final speed of Y after X has travelled 3.0 m along the plane?

- A 1.6 m s^{-1}
- B 2.3 m s^{-1}
- C 3.3 m s^{-1}
- D 4.3 m s^{-1}

10 A mass m is attached to one end of an elastic string of an unstretched length l . When the